

OpenClaw × □□□□□□□□□□□□□□

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OpenClaw

OpenClaw

1.1 背景

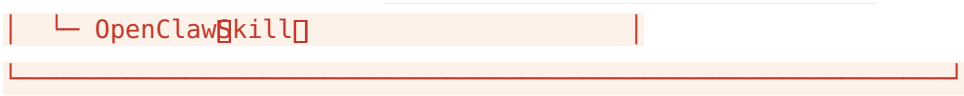
1990年「Open Complex Giant System, OCGS」**OpenClaw**

問題	OpenClaw問題	期待される結果
問題1	OpenClawのAPIを呼び出す	成功/失敗/エラー
問題2	AgentのSkillを実行する	実行結果
問題3	環境の状態を確認する	現在の状態
問題4	Agent→Skill→環境	動作確認
問題5	環境の初期化	1+1>2を確認

1.2 "OpenClaw"

OpenClaw

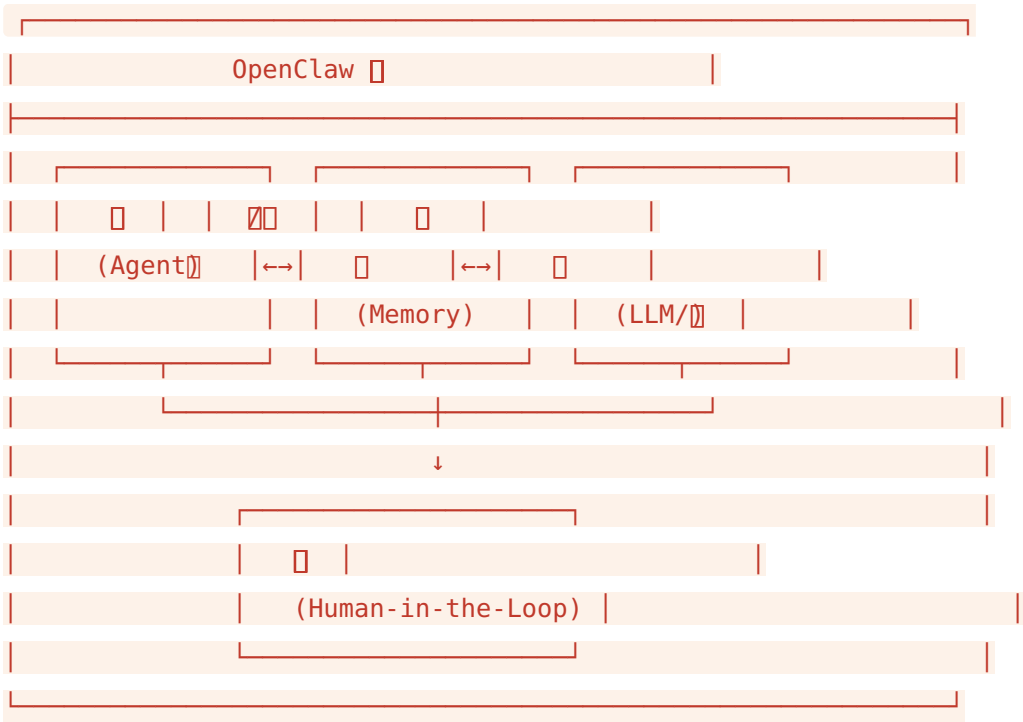
```
graph TD; Systematics -->leftrightarrow;leftrightarrow-->Systematology;leftrightarrow-->OpenClaw;Systematology-->"+ +";"+ +"-->Agent;Agent-->Systems_Engineering[Systems Engineering];
```



[REDACTED] OpenClaw [REDACTED]

2.1 "[REDACTED]" [REDACTED] HWME [REDACTED]

[REDACTED]1992 [REDACTED]" [REDACTED]" OpenClaw [REDACTED]



[REDACTED] OpenClaw [REDACTED]

1. [REDACTED] Agent [REDACTED]

- [REDACTED] Agent [REDACTED]
- [REDACTED] Agent Alpha [REDACTED] Bravo [REDACTED] Charlie [REDACTED] Delta [REDACTED] Echo [REDACTED]
- [REDACTED] Agent [REDACTED] Agent

2. [REDACTED] Memory [REDACTED]

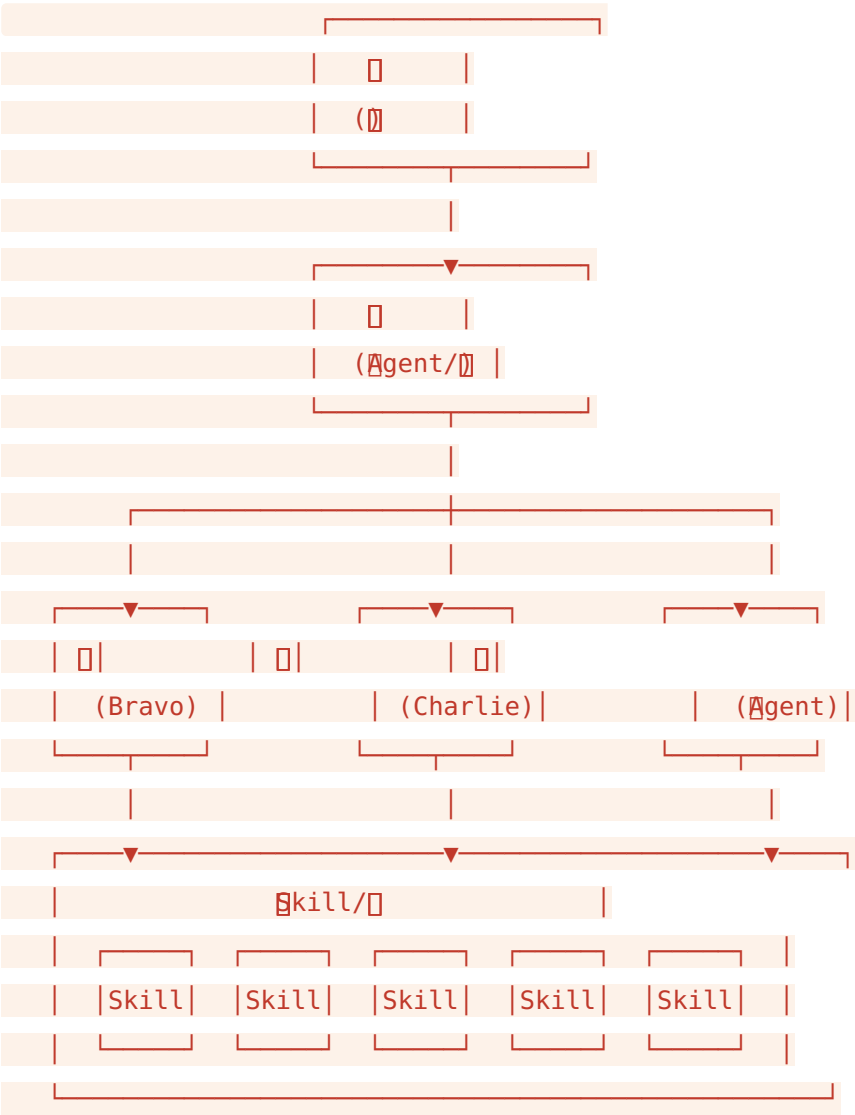
- Active Memory [REDACTED]
- Short-term Memory [REDACTED] memory/YYYY-MM-DD.md [REDACTED]
- Long-term Memory [REDACTED] MEMORY.md [REDACTED]
- External Memory [REDACTED] GitHub [REDACTED]
- Vector Memory [REDACTED]

3. [redacted] Tool [redacted]

- LLM [redacted]
- Skill [redacted]
- [redacted] API [redacted]

2.2 "[redacted]" OpenClaw [redacted]

[redacted]"[redacted]" [redacted] OpenClaw [redacted]



[redacted]

[redacted] Technical Authority [redacted]

- [redacted]
- [redacted] Bravo Agent + [redacted]
- [redacted]"[redacted]"

Administrative Authority

- [] [] [] [] [] [] [] []
- [][]Charlie Agent + [][][][]
- [][]"[][][][][][]"

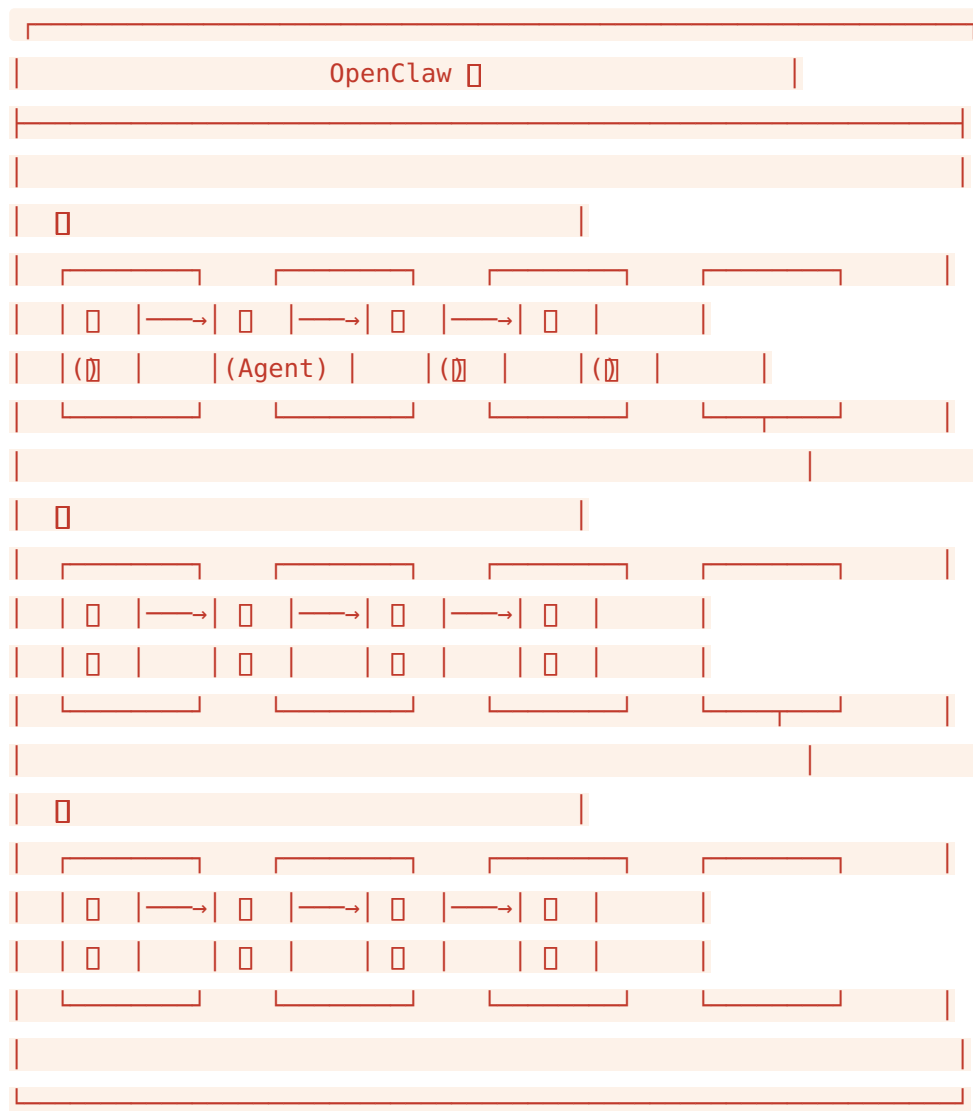
□□□□□□Technical Committee□

- `XXXXXXXXXXXXXXXXXXXX`
- `XXXXAgentXX + XXXXXX`
- `XXX"XXXXXXXXXX"`

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3.1 OpenClaw " " "

OpenClaw



클래스 정의

1. TaskFeedbackController

```
// 클래스 정의
class TaskFeedbackController {
    async execute(task) {
        // 예측
        const prediction = await this.predict(task);

        // 실행
        const result = await this.agent.run(task);

        // 평가
        const error = this.evaluate(result, prediction);
        if (error > threshold) {
            await this.correct(task, result, error);
        }

        // 학습
        await this.learn(task, result, error);
    }
}
```

2. MemoryFeedback

```
// 클래스 정의
class MemoryFeedback {
    async retrieve(query) {
        // 검색
        const results = await this.memory.search(query);

        // 관련성 평가
        const relevance = await this.getUserFeedback(results);

        // 조정
        await this.adjustWeights(query, results, relevance);

        // 업데이트
        await this.updateIndex();
    }
}
```

```
}  
}
```

3.

```
//     
class SystemHealthFeedback {  
  async monitor() {  
    //     
    const metrics = await this.collectMetrics();  
  
    //     
    const anomalies = this.detectAnomalies(metrics);  
  
    //     
    if (anomalies.length > 0) {  
      await this.autoAdjust(anomalies);  
    }  
  
    //     
    await this.generateReport(metrics, anomalies);  
  }  
}
```

3.2 OpenClaw " "

 Robustness —

OpenClaw

1.

```
    
|    |  
| -    |  
| -    |  
    
|    |  
| -    |  
| -    |  
  
```

- Agent	
-	

2. graceful degradation

```
class RobustExecution {
  async executeWithFallback(task) {
    const strategies = [
      // 
      () => this.fullExecution(task),
      // 
      () => this.simplifiedExecution(task),
      // 
      () => this.basicExecution(task),
      // 
      () => this.errorResponse(task)
    ];

    for (const strategy of strategies) {
      try {
        return await strategy();
      } catch (error) {
        console.warn(` ${error.message}`);
        continue;
      }
    }
  }
}
```

3.

```
class AdaptiveController {
  async adjustParameters() {
    // 
    const state = await this.getState();

    // PID
    const error = state.target - state.current;
```



```
const adjustment = this.pidController.calculate(error);  
  
//   
await this.applyAdjustment(adjustment);  
  
//   
await this.logAdjustment(state, adjustment);  
}  
}
```

  " "  

4.1

  " "  OpenClaw 

```
┌───────────────────────────────────────────────────────────────────────────────────┐  
│ OpenClaw   │  
└───────────────────────────────────────────────────────────────────────────────────┘  
┌───────────────────────────────────────────────────────────────────────────────────┐  
│  GitHub/GitLab  │  
│ └   │  
│ └  + GitHub Actions  │  
│ └    │  
└───────────────────────────────────────────────────────────────────────────────────┘  
│   │  
│ └ MEMORY.md   curated wisdom  │  
│ └ memory/*.md   raw logs  │  
│ └ SQLite   semantic search  │  
│ └    │  
└───────────────────────────────────────────────────────────────────────────────────┘  
│   │  
│ └ BM25   │  
│ └ Vector   │  
│ └ Temporal Decay   │  
│ └ MMR   │  
└───────────────────────────────────────────────────────────────────────────────────┘  
│    │  
└───────────────────────────────────────────────────────────────────────────────────┘
```



4.2 "Memory Consolidation"

"Memory Consolidation" [REDACTED]

1. [REDACTED] vs [REDACTED]

```

class HybridMemory {
  // [REDACTED]
  async passiveRecord(event) {
    await this.writeToSessionLog(event);
    await this.updateVectorIndex(event);
  }

  // [REDACTED]
  async activeMark(content, importance) {
    await this.promoteToLongTerm(content);
    await this.setImportanceScore(content, importance);
    await this.triggerConsolidation();
  }
}

```

2. [REDACTED] Memory Consolidation

```

class MemoryConsolidation {
  // [REDACTED]
  async consolidate() {
    // 1. [REDACTED]
    const recentMemories = await this.getRecentMemories(days=7);

    // 2. [REDACTED]
    const keyInsights = await this.identifyKeyInsights(recentMemories);

    // 3. [REDACTED]
    for (const insight of keyInsights) {
      await this.updateLongTermMemory(insight);
    }
  }
}

```

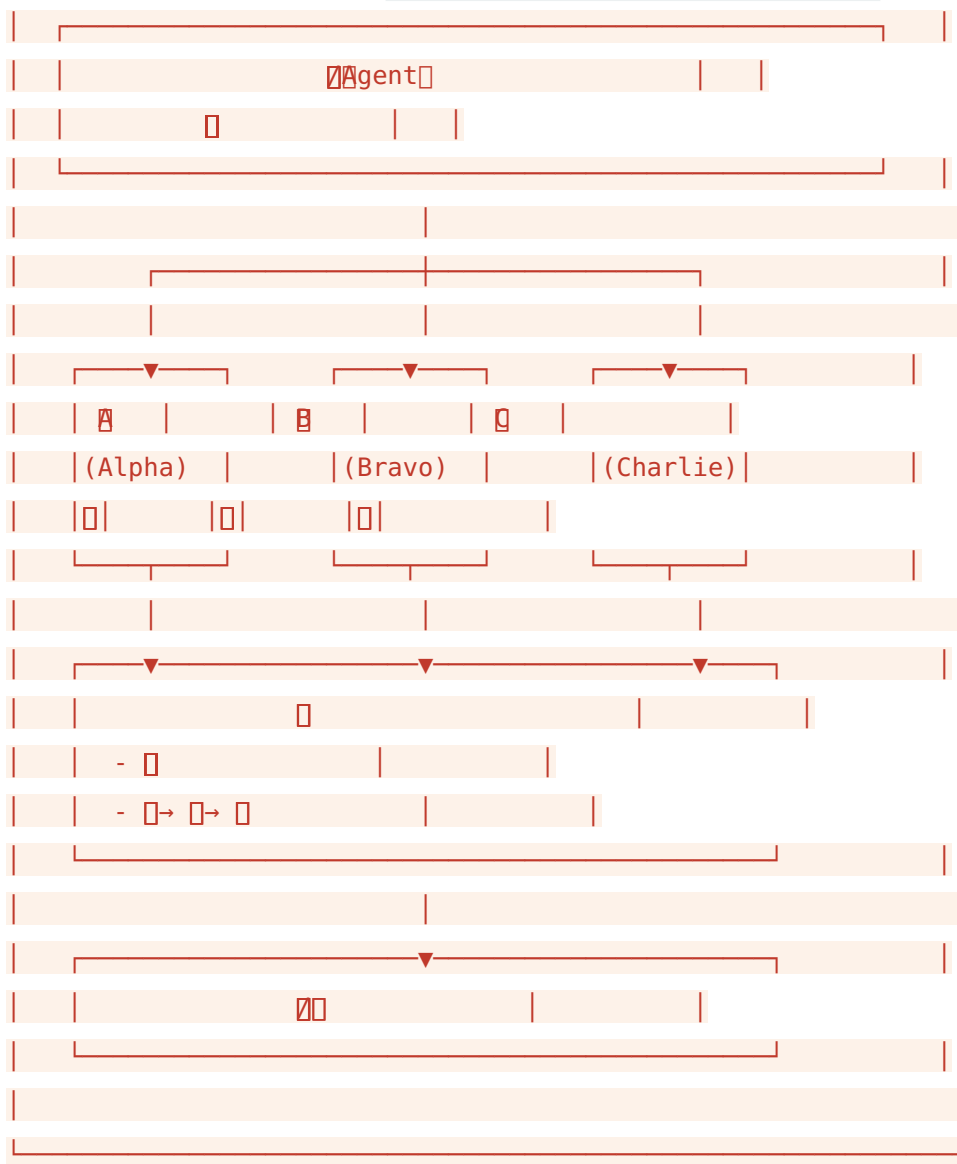
```
class MetaSynthesisRetrieval {
  async retrieve(query) {
    // 1. □
    const keywordResults = await this.bm25Search(query);
    const semanticResults = await this.vectorSearch(query);
    const temporalResults = await this.temporalSearch(query);

    // 2. □
    const synthesized = await this.synthesize({
      keyword: keywordResults,
      semantic: semanticResults,
      temporal: temporalResults
    });

    // 3. □LM□
    const ranked = await this.expertRank(synthesized, query);

    return ranked;
  }
}
```

OpenCLaw (HWME)	
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5.2

```
class HWMEProcess {  
  async conductWorkshop(task) {  
    // Phase 1:   
    const expertOpinions = await this.gatherExpertOpinions(task);  
  
    // Phase 2:   
    const quantitativeAnalysis = await this.quantitativeAnalysis(expertOpinions);  
  
    // Phase 3:   
    const synthesis = await this.metaSynthesis({  
      opinions: expertOpinions,  
      analysis: quantitativeAnalysis
```

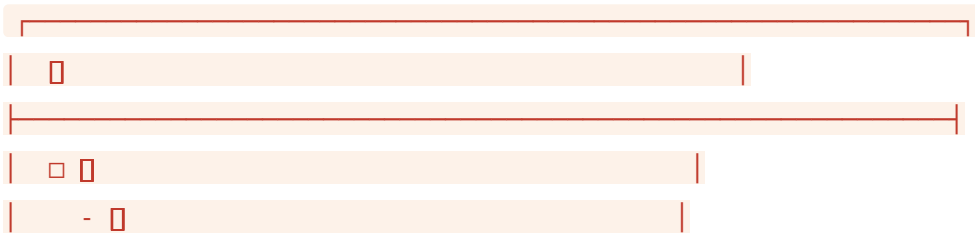
```
});  
  
// Phase 4:   
const decision = await this.expertDecision(synthesis);  
  
// Phase 5:   
await this.learnFromOutcome(task, decision);  
  
return decision;  
}  
}
```

5.3 Agent

Agent			
Alpha			
Bravo			
Charlie		Heartbeat	
Delta			
Echo			
Foxtrot			

"

1-3



[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] 3-6 [REDACTED]

[REDACTED]"[REDACTED]"Agent[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] Agent[REDACTED]

[REDACTED] - Agent[REDACTED]

[REDACTED] - Agent[REDACTED]

[REDACTED] [REDACTED]

[REDACTED] - Agent[REDACTED]

[REDACTED] - [REDACTED]

[REDACTED] [REDACTED]

[REDACTED] - Bravo[REDACTED]

[REDACTED] - Charlie[REDACTED]

[REDACTED] - Agent[REDACTED]

[REDACTED]

[REDACTED] 6-12 [REDACTED]

[REDACTED]"[REDACTED]"——[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] [REDACTED]

[REDACTED] - [REDACTED]

[REDACTED] - [REDACTED]

[REDACTED] [REDACTED]



[redacted]"[redacted]"[redacted]

7.1 [redacted]

[redacted]	[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	>99%	[redacted]/[redacted]
	[redacted]	<30s	[redacted]
[redacted]	[redacted]	<1s	[redacted]
	[redacted]	>85%	[redacted]vs[redacted]
[redacted]	[redacted]	>90%	[redacted]/[redacted]
	[redacted]	>70%	[redacted]/[redacted]

7.2 [redacted]

[redacted]	[redacted]	[redacted]	[redacted]
Agent[redacted]	[Agent[redacted]]	>40%	[Agent[redacted]/[redacted]
	[redacted]	>95%	[redacted]/[redacted]
[redacted]	[redacted]	10-20%	[redacted]/[redacted]
	[redacted]	>4.5/5	[redacted]
[redacted]	[redacted]	>80%	[redacted]/[redacted]

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[illegible][illegible]

1. 
2. 
3. 
4. 
5. 

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"

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2026年5月9日

OpenClaw